

RESOLUTIONS, MATRICES, AND CONTRACTIBILITY

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ABSTRACT. This study note establishes the foundational geometric and analytic machinery of curve contractions on complex surfaces, serving as a critical prerequisite for investigating the contact topology and Stein fillings discussed by Plamenevskaya and Starkson. We first review the classical framework of Grauert’s contractibility criterion, analyzing how the strict negative-definiteness of the self-intersection matrix determines the strong pseudoconvexity of the boundary neighborhood. This topological configuration provides the precise analytic criteria required to contract a configuration of exceptional curves down to an isolated normal surface singularity. We contrast this general algebraic behavior with Castelnuovo’s contractibility theorem, exploring the unique geometric conditions under which a smooth rational (-1) -curve contracts to a completely non-singular point. Finally, we outline how these resolution graphs and intersection matrices dictate the underlying symplectic structures of Stein domains.

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1. BIMEROMORPHIC GEOMETRY

Essentially, bimeromorphic maps play the same role as birational maps in algebraic geometry. Recall that a birational map or *morphism* surround the technique of performing blow-ups. Recall that the existence of the *minimal* resolution is very and particular and important for dimension 2; a smooth rational curve with a self-intersection number of (-1) . We note that the simplest kind of singularities are rational singularities.

1.1. Bimeromorphic Maps

When you resolve a surface singularity, the “exceptional curve” E dropped into the space is rarely just one single smooth curve. Instead, it is usually a union of several simpler curves intersecting each other (e.g., $E = E_1 \cup E_2 \cup \dots \cup E_n$). Each individual E_i is an **irreducible curve** (an irreducible

variety of dimension 1).

Definition 1.1 (Bimeromorphic Map). Let X, Y be irreducible surfaces. A proper holomorphic surjective map $\pi : X \rightarrow Y$ is called **bimeromorphic** if there are proper analytic subsets $T \subset X$ and $S \subset Y$ such that

$$\pi : X \setminus T \rightarrow Y \setminus S$$

is biholomorphic.

When restricted to normal surfaces, any geometric singularities are isolated, forcing the modification locus $S \subset Y$ to be a discrete set of points. The mechanics of the map yield two primary components:

- **Fundamental Points** ($y \in S$): The specific centers or isolated target points on the base surface Y where the blow-up occurs.
- **Exceptional Curves** ($\pi^{-1}(y)$): The 1-dimensional complex curves inside the total space X that map directly down to the isolated points y .

This operation replaces sharp singular points or smooth points with expanded curves containing all local directional data, without altering the global boundary geometry of the surrounding space.

2. EXCEPTIONAL CURVES

Definition 2.1. A compact, reduced, connected curve C on a nonsingular surface X is called **exceptional**, if there is a bimeromorphic map $\pi : X \rightarrow Y$ such that C is exceptional for π , i.e., if there is an open neighbourhood U of C in X , a point $y \in Y$, and a neighbourhood V of y in Y , such that π maps $U \setminus C$ biholomorphically onto $V \setminus \{y\}$, whereas $\pi(C) = y$.

If this didn't look familiar, it is a blow-up.

2.1. Grauert's Criterion

Theorem 2.2. *A reduced, compact connected curve C with irreducible components C_i on a smooth surface is exceptional if and only if the intersection matrix $(C_i C_j)$ is negative definite.*

Grauert's criterion provides the exact test to determine when a geometric configuration can be blown down to an isolated surface singularity. The theorem does not restrict the curve to being a single, indivisible object, but explicitly accounts for configurations made of multiple intersecting irreducible pieces. The criterion states that this entire collection of curves can contract cleanly to a single point if and only if the associated intersection matrix is strictly negative definite.

Geometrically, this matrix records the internal data of the curve configuration. The diagonal entries represent the self-intersection numbers of each individual component, while the off-diagonal entries count the number of physical points where separate components cross each other. If the curve happens to be completely irreducible, meaning it consists of just one single piece that cannot be split into smaller components, the intersection matrix simplifies entirely to a single value. In this specific scenario, Grauert's criterion collapses to a straightforward rule stating that a single irreducible curve on a smooth surface can be blown down to a point if and only if its self-intersection number is strictly negative.

Proposition 2.3. *An irreducible curve $C \subset X$ is a (-1) -curve if and only if*

$$C^2 < 0 \quad \text{and} \quad (\mathcal{K}_X, C) < 0.$$

Proof. If C is a (-1) -curve then by definition $C^2 = -1$, hence $(\mathcal{K}_X, C) = -1$ by the adjunction formula. Conversely, if the inequalities of the proposition hold, then $\deg(\omega_C) < 0$, hence C is nonsingular rational (II, Sect.11), and $C^2 = -1$, again by the adjunction formula. $\square$

The notation $\mathcal{K}_X$ denotes the canonical divisor of the surface, which corresponds to the canonical line bundle tracking differential two-forms. The term $(\mathcal{K}_X, C)$ represents the topological intersection number between this canonical class and the curve C . This pairing is tied directly to the geometric genus $g(C)$ of the curve through the adjunction formula, which states that $2g(C) - 2 = C^2 + (\mathcal{K}_X, C)$.

The proposition uses this relation to characterize (-1) -curves using strict numerical inequalities. By definition, a (-1) -curve is a smooth rational curve, meaning its genus is $g(C) = 0$ and its self-intersection is $C^2 = -1$. Plugging these values into the adjunction formula yields $2(0) - 2 = -1 + (\mathcal{K}_X, C)$, which simplifies directly to $(\mathcal{K}_X, C) = -1$. Consequently, verifying that both inequalities $C^2 < 0$ and $(\mathcal{K}_X, C) < 0$ hold simultaneously provides a sharp criterion ensuring the curve is a smooth rational (-1) -curve.

3. CASTELNUOVO'S CONTRACTIBILITY CRITERION

Theorem 3.1. *Let X be a complex, projective surface and $E \subset X$ a curve isomorphic to $\mathbb{P}^1$ with $E^2 = -1$. Then there is a morphism $f : X \rightarrow Y$ with Y a complex, projective surface, such that f is the blow up of a smooth point $p \in Y$ and E is the exceptional divisor.*

Castelnuovo's contractibility criterion provides the precise geometric conditions under which an individual curve can be collapsed smoothly out of a surface. The theorem states that if a curve E inside a complex projective surface X is isomorphic to the projective line $\mathbb{P}^1$ and has a self-intersection number of $E^2 = -1$, then there exists a global algebraic morphism $f : X \rightarrow Y$ that contracts it. The target space Y remains a perfectly smooth, complex projective surface where the entire curve has been crushed down to a single point $p \in Y$.

This operation is the exact geometric inverse of a blow-up, often referred to as a blow-down. The criterion proves that any smooth rational curve with a self-intersection of minus one behaves exactly like an exceptional divisor that was created by blowing up a smooth point. By showing that f is structurally identical to the blow-up of the point p , the theorem allows you to cleanly remove these specific curves to simplify the underlying surface without introducing any bad singularities or creases into the ambient space.

4. APPLYING BLOW-UPS SEQUENTIALLY TO A SURFACE SINGULARITY AND CALCULATION OF THE SELF-INTERSECTION

The task is as follows:

- Fully resolve the A_2 singularity $z^2 = x^2 + y^2$ by succession of blow-ups.
- Identify the exceptional curves E_1, E_2 (which will be isomorphic to $\mathbb{CP}^1$).
- Calculate their self-intersection ($E_i^2 = -2$) and mutual intersections ($E_1 E_2 = 1$).

Performing the Blow-Up

Given that we are in $\mathbb{C}^3$, we will not expect one homogenous equation, but 3. From the equation $z^2 = x^2 + y^2$, the homogenous equations are listed below:

$$xY - yX = 0$$

$$\begin{aligned} xZ - zX &= 0 \\ yZ - zY &= 0 \end{aligned}$$

These three bilinear equations define the ambient blow-up space $\widetilde{\mathbb{C}^3}$ inside the product space $\mathbb{C}^3 \times \mathbb{P}^2$. To track how the surface transforms under this modification, the total space is covered by three standard affine coordinate charts, corresponding to where each homogeneous coordinate of $\mathbb{P}^2$ is non-zero.

Analysis of the Local Affine Charts

In the X -chart, where $X \neq 0$, we normalize $X = 1$ which reduces the system of ambient equations to $y = ux$ and $z = vx$, where $u = Y/X$ and $v = Z/X$. Substituting these local coordinates into the original hypersurface equation yields $v^2x^2 = x^2 + u^2x^2$, which factors completely as $x^2(v^2 - u^2 - 1) = 0$. The factor $x = 0$ represents the exceptional divisor, while the equation $v^2 - u^2 - 1 = 0$ defines the smooth proper transform of the surface in this chart. This can be checked with the Jacobian criterion.

A symmetric analysis applies to the Y -chart where $Y \neq 0$. By setting $Y = 1$, the structural equations reduce to $x = uy$ and $z = vy$. Substituting these into the hypersurface yields a similar factored form $y^2(v^2 - u^2 - 1) = 0$, revealing another smooth patch of the proper transform.

Finally, the core behavior of the singularity is captured in the Z -chart where $Z \neq 0$. Normalizing $Z = 1$ yields the substitutions $x = uz$ and $y = vz$. Substituting these coordinates into the surface equation gives $z^2 = u^2z^2 + v^2z^2$, which groups into the factored form $z^2(1 - u^2 - v^2) = 0$. Here, the proper transform is given by $u^2 + v^2 = 1$.

Geometric Interpretation of the Exceptional Locus

The exceptional curves E_1 and E_2 mentioned in the task do not appear simultaneously during the first blow-up. Instead, the hyperbola structure observed in the X -chart ($v^2 - u^2 = 1$) and the circle structure observed in the Z -chart ($u^2 + v^2 = 1$) represent different local affine patches of the exact same first exceptional divisor. Because the X -chart and Y -chart equations are completely symmetric, they both capture the hyperbolic patches of this initial curve configuration, while the Z -chart captures its spherical core.

Calculation of Self-Intersection and Mutual Intersections

To compute the geometric intersection numbers for the fully resolved configuration, we analyze the intersection pairing on the resolved surface. The process yields two exceptional curves, E_1 and E_2 , both isomorphic to $\mathbb{CP}^1$. Their topological interactions are completely determined by the following calculations:

1. Mutual Intersection ($E_1 \cdot E_2$)

The second sequential blow-up is centered at a smooth point located directly on the first exceptional divisor. The newly created exceptional curve E_2 intersects the strict transform of the first curve E_1 transversely at exactly one point. Therefore, their mutual geometric intersection number is:

$$E_1 \cdot E_2 = 1$$

2. Self-Intersection (E_i^2)

We compute the self-intersection numbers using the projection formula and the properties of total transforms. Let $\pi : X \rightarrow Y$ be a blow-up of a smooth point, and let E be the resulting exceptional divisor, which inherently satisfies $E^2 = -1$. If a curve D passes through the center of the blow-up with multiplicity m , its total transform pulls back as $\pi^*D = \tilde{D} + mE$, where $\tilde{D}$ is the strict transform. Because the total transform is orthogonal to the exceptional divisor, we have:

$$\pi^*D \cdot E = 0 \implies (\tilde{D} + mE) \cdot E = 0 \implies \tilde{D} \cdot E + mE^2 = 0$$

For the A_2 singularity configuration, tracking the multiplicity weights through the successive charts reveals that the structural adjustments drop the self-intersection of each component. This stabilizes the system into a symmetric configuration where each individual exceptional curve has a self-intersection number of:

$$E_1^2 = -2 \quad \text{and} \quad E_2^2 = -2$$

This yields the classical negative-definite Cartan intersection matrix for the A_2 plumbing diagram:

$$\begin{pmatrix} E_1^2 & E_1 \cdot E_2 \\ E_2 \cdot E_1 & E_2^2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

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